Qualifying Exam Syllabus

Name: Songyu Ye
Date: May 14, 2026
Time: 9am - 12pm
Location: Evans 732
Committee: Constantin Teleman (advisor), David Nadler (exam chair), Hannah Larson, Richard Borcherds

1. Major Topic: Representation Theory (Algebra)

- **Lie algebras:** solvable, nilpotent, and semisimple Lie algebras; Cartan subalgebras; root systems; Weyl groups; Dynkin diagrams; classification of simple Lie algebras; Killing form; complete reducibility; examples: $\mathfrak{sl}_n$, $\mathfrak{so}_n$, $\mathfrak{sp}_{2n}$, $\mathfrak{u}_n$.
- **Lie groups and algebraic groups:** fundamental group and center of Lie groups; simply connected and adjoint forms associated to a simple Lie algebra; real forms, compact forms, split forms; Cartan involution; maximal compact subgroups of real Lie groups; reductive, Borel subgroups, parabolic subgroups, Levi subgroups.
- **Representation theory:** universal enveloping algebra; PBW theorem; Verma modules; highest weight modules; Casimir elements; central characters; Weyl character formula.

Reference: Kirillov, *Introduction to Lie Groups and Lie Algebras*. Varadarajan, *Lie Groups, Lie Algebras, and Their Representations*.

2. Major Topic: Algebraic Geometry (Algebra)

- **Sheaves:** presheaves and sheaves; morphisms of sheaves; stalks; exact sequences of sheaves; skyscraper sheaves; locally free sheaves; quasi-coherent sheaves.
- **Morphisms of schemes:** affine morphisms; open and closed immersions; finite and integral morphisms; separated, proper, and projective morphisms; flat, smooth, unramified, and étale morphisms; fiber products and base change.
- **Divisors and bundles:** Cartier divisors and Weil divisors; linear equivalence; Picard groups; linear systems; line bundles; ampleness and very ampleness; vector bundles; Chern classes; normal bundles of subvarieties; adjunction formula; sheaf of differentials.
- **Sheaf cohomology and derived functors:** cohomology of sheaves; cohomology of projective spaces; pullback and pushforward; higher direct images; vanishing theorems; Serre duality; flatness; cohomology and base change.
- **Curves:** Riemann–Roch; Riemann–Hurwitz; curves in projective spaces; elliptic curves; hyperelliptic curves.

Reference: Hartshorne, *Algebraic Geometry*, Chapters 2–4.

3. Minor Topic: Algebraic Topology (Geometry)

- **Fundamental groups and homotopy:** Fundamental group; Van Kampen's theorem; fundamental groups of graphs and surfaces; covering space theory; universal covers; deck transformations; CW complexes; higher homotopy groups; Whitehead's theorem; Hurewicz theorem.
- **Homology and cohomology:** cellular and simplicial homology; de Rham cohomology; long exact sequences; Mayer–Vietoris sequence; relative homology and cohomology; excision; universal coefficient theorem; Künneth theorem; cup product in cohomology; Poincaré duality for closed orientable manifolds;

Reference: Hatcher, *Algebraic Topology*, Chapters 1–4, excluding additional topics.